

Alignment + Accuracy: The Cascade Reward Representation for Preranking

Supplementary Material

Hedi Xia
Pinterest
San Francisco, CA, USA
hxia@pinterest.com

Yichu Zhou
Pinterest
Seattle, WA, USA
yichuzhou@pinterest.com

Bella Huang
Pinterest
San Francisco, CA, USA
shuang@pinterest.com

Dafang He
Pinterest
San Francisco, CA, USA
dhe@pinterest.com

Dylan Zhou
Pinterest
San Francisco, CA, USA
shenglangzhou@pinterest.com

Zili Li
Pinterest
San Francisco, CA, USA
zilili@pinterest.com

Hongbo Deng
Pinterest
San Francisco, CA, USA
hdeng@pinterest.com

Darren Reger
Pinterest
San Francisco, CA, USA
dreger@pinterest.com

James Li
Pinterest
San Francisco, CA, USA
jamesli@pinterest.com

Yali Bian
Pinterest
San Francisco, CA, USA
ybian@pinterest.com

Tianyou Wang
Pinterest
San Francisco, CA, USA
tianyouwang@pinterest.com

Piyush Maheshwari
Pinterest
San Francisco, CA, USA
pmaheshwari@pinterest.com

Bowen Deng
Pinterest
San Francisco, CA, USA
bdeng@pinterest.com

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This document provides supplementary material for the main paper. Section numbers and theorem references follow the main text.

A Proofs for the Cascade Reward Representation (Section 3)

This appendix provides detailed proofs for the Cascade Reward Representation (Theorem 3.1) and its supporting lemmas. All results use the non-increasing impression profile [3] and the three-stage

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pipeline model from the main text. We restore the user subscript u where needed because the key step—bounding the remainder—requires averaging across users. The four steps of the main-text proof sketch correspond to the four subsections below: Step 1 to §A.1 (prefix-average reward expansion), Step 3 to §A.2 (overlap-shift alignment expansion), Step 4 to §A.3 (full-pool to logged-overlap bridge) and §A.4 (impressed-accuracy estimator consistency), and the combined statement to §A.5.

Figure 1 summarizes the dependency structure of the appendix results and how each maps back to a claim in the main text.

Worked example: reward lift through candidate forwarding

Example A.1. Let $D = \{a, b, c, d, e\}$ with $k=3$, impression probabilities $I(1)=1.0$, $I(2)=0.8$, $I(3)=0.5$, and engagement probabilities $E(a)=0.3$, $E(b)=0.5$, $E(c)=0.6$, $E(d)=0.1$, $E(e)=0.2$. *Preranker A* selects $C = \{a, c, d\}$; L_2 orders them c, a, d , giving $R = 0.6 \times 1.0 + 0.3 \times 0.8 + 0.1 \times 0.5 = 0.89$. *Preranker B* selects $C' = \{b, c, e\}$; L_2 orders them c, b, e , giving $R' = 0.6 \times 1.0 + 0.5 \times 0.8 + 0.2 \times 0.5 = 1.10$. The reward lift of $+0.21$ arises solely from *Preranker B* forwarding a stronger set to L_2 , illustrating the dependence the decomposition formalizes.

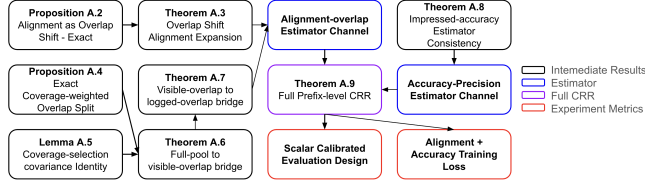


Figure 1: Proof-dependency graph of the theory appendix. The alignment path combines the overlap-shift expansion with the full-pool-to-logged-overlap bridge to obtain the alignment-overlap estimator channel. The accuracy path uses impressed labels to obtain the accuracy-precision estimator channel. The two channels combine in the full prefix-level CRR, which motivates the scalar calibrated evaluation metric and the two-branch alignment+accuracy training loss used in experiments.

A.1 Prefix-average expansion of served reward

Equation (3) in the main text writes the user-level reward as a weighted sum of prefix-average engagements. We derive it via Abel’s summation (summation by parts).

DERIVATION OF (3). Let $a_i := E_u(s_i)$ and $b_i := I_u(i)$, and define the partial sums $A_p := \sum_{i=1}^p a_i = p S_{p,u}$. With the convention $I_u(k+1) := 0$, Abel’s summation gives:

$$\begin{aligned} R_u &= \sum_{i=1}^k a_i b_i = A_k b_k + \sum_{p=1}^{k-1} A_p (b_p - b_{p+1}) \\ &= \sum_{p=1}^k p S_{p,u} (I_u(p) - I_u(p+1)) = \sum_{p=1}^k w_{p,u} S_{p,u}, \end{aligned}$$

where $w_{p,u} := p(I_u(p) - I_u(p+1)) \geq 0$ because the impression profile is non-increasing. $\square$

A.2 Alignment as overlap shift times boundary engagement

We first establish an exact identity for the alignment channel at a fixed prefix p and user u , using the simplified three-stage pipeline (L_0, L_1, L_2) from the main text [1, 2, 8]. Define the overlap count $\hat{p}_u := |C_u^E \cap X_{p,u}^0|$ and the overlap shift $\Delta_{p,u} := \hat{p}_u - p$.

Define the *boundary average* $\partial S_{p,u}$: the average engagement of items between positions p and $\hat{p}_u$ in the L_2 ranking:

$$\partial S_{p,u} := \begin{cases} \frac{1}{\hat{p}_u - p} \sum_{i=p+1}^{\hat{p}_u} E_u(s_i), & \hat{p}_u > p, \\ \frac{1}{p - \hat{p}_u} \sum_{i=\hat{p}_u+1}^p E_u(s_i), & \hat{p}_u < p, \\ 0, & \hat{p}_u = p. \end{cases}$$

PROPOSITION A.2 (ALIGNMENT AS OVERLAP SHIFT – EXACT). For every prefix p and every candidate set C_u , with $\hat{p}_u = |C_u \cap X_{p,u}^0|$ and $\Delta_{p,u} := \hat{p}_u - p$:

$$\mathcal{A}_{p,u}(C_u) = -\frac{\Delta_{p,u}}{p} \partial S_{p,u},$$

with the convention that $\mathcal{A}_{p,u} = 0$ when $\hat{p}_u = p$.

PROOF. Since L_2 induces a strict total order [7], items in $C_u \cap X_{p,u}^0$ occupy positions $1, \dots, \hat{p}_u$ in the L_2 ranking of C_u . Thus $p S_{p,u}^{\text{thr}} = \sum_{i=1}^{\hat{p}_u} E_u(s_i)$.

$$\text{Case } \hat{p}_u > p. \text{ Split: } p S_{p,u}^{\text{thr}} = \underbrace{\sum_{i=1}^p E_u(s_i)}_{= p S_{p,u}} + \underbrace{\sum_{i=p+1}^{\hat{p}_u} E_u(s_i)}_{= (\hat{p}_u - p) \partial S_{p,u}}. \text{ Hence}$$

$$S_{p,u}^{\text{thr}} = S_{p,u} + \frac{\hat{p}_u - p}{p} \partial S_{p,u}, \text{ and}$$

$$\mathcal{A}_{p,u} = S_{p,u} - S_{p,u}^{\text{thr}} = -\frac{\hat{p}_u - p}{p} \partial S_{p,u} = -\frac{\Delta_{p,u}}{p} \partial S_{p,u}.$$

Case $\hat{p}_u < p$. Split

$$p S_{p,u} = \sum_{i=1}^{\hat{p}_u} E_u(s_i) = p S_{p,u}^{\text{thr}} + (p - \hat{p}_u) \partial S_{p,u}.$$

Hence $S_{p,u}^{\text{thr}} = S_{p,u} - \frac{p - \hat{p}_u}{p} \partial S_{p,u}$, and therefore

$$\mathcal{A}_{p,u} = \frac{p - \hat{p}_u}{p} \partial S_{p,u} = -\frac{\Delta_{p,u}}{p} \partial S_{p,u}.$$

Case $\hat{p}_u = p$. $\Delta_{p,u} = 0$, so $\mathcal{A}_{p,u} = 0$. $\square$

A.2.1 Population first-order expansion of the alignment channel. The exact identity in Proposition A.2 expresses $\mathcal{A}_{p,u}$ as a product of $\Delta_{p,u}$ and a bounded coefficient:

$$\mathcal{A}_{p,u} = c_{p,u} \Delta_{p,u}, \quad c_{p,u} := -\frac{\partial S_{p,u}}{p}. \quad (1)$$

The “first-order expansion” replaces $c_{p,u}$ —which depends on the experimental candidate set through the boundary average $\partial S_{p,u}$ —by its baseline-threshold component plus a controlled residual that vanishes in expectation across users.

Assumptions for the alignment expansion. The expansion below requires two regularity conditions.

ASSUMPTION 1 (Bounded engagements). For all users u and items s_i : $E_u(s_i) \in [0, 1]$.

This holds by definition since engagement probabilities (click, save, etc.) lie in $[0, 1]$. Under this assumption, the per-user coefficient satisfies $c_{p,u} \in [-1/p, 0]$, and $|B_u| \leq 1$ for the boundary deviation defined below.

ASSUMPTION 2 (Conditional population boundary smoothness). Let $B_u := \partial S_{p,u} - E_u(s_{p,u}^0)$ be the boundary deviation defined in (2) below. For each overlap shift d , the conditional boundary mean satisfies

$$|\mathbb{E}_u[B_u \mid \Delta_{p,u} = d]| \leq M |d|$$

for some constant $M > 0$.

This is the population-level smoothness actually needed by the proof. It does not require adjacent items to have similar engagement for a single user. Instead, after grouping users by the size of the overlap shift, the mean engagement of the boundary items is allowed to drift from the baseline threshold item only linearly in that shift. Unconditional smoothness of the population engagement curve is a useful diagnostic, but the proof uses this conditional version because $\Delta_{p,u}$ is itself selected by the model comparison.

Per-user split of the boundary-engagement coefficient. Let $s_{p,u}^0$ denote the item at position p in L_2 's ranking of the baseline candidate set C_u^0 . Define the *boundary deviation*:

$$B_u := \partial S_{p,u} - \mathbb{E}_u(s_{p,u}^0). \quad (2)$$

Then from (1):

$$\mathcal{A}_{p,u} = \underbrace{-\frac{\mathbb{E}_u(s_{p,u}^0)}{p} \Delta_{p,u}}_{\text{leading term}} - \underbrace{\frac{B_u}{p} \Delta_{p,u}}_{\text{remainder } R_{p,u}}. \quad (3)$$

The leading coefficient $-\mathbb{E}_u(s_{p,u}^0)/p$ depends only on the baseline setup and the user, not on the experimental preranker.

Why pointwise Lipschitz fails. To bound $R_{p,u}$ for a single user, one would need $|B_u| = O(|\Delta_{p,u}|)$, which requires $|\mathbb{E}_u(s_i) - \mathbb{E}_u(s_j)| \leq M|i - j|$ for items near position p . This is unrealistically strong: the main ranker L_2 orders items by *score*, not engagement, so adjacent positions can have very different engagements for a single user. Assumption 2 avoids this by requiring smoothness only of the *population conditional mean*, not of individual users' engagement curves.

Boundary bias-noise decomposition. Define $\bar{B}(d) := \mathbb{E}_u[B_u \mid \Delta_{p,u} = d]$, the average boundary deviation among users with overlap shift d . Assumption 2 gives the bound

$$|\bar{B}(d)| \leq M|d|. \quad (4)$$

Decompose $B_u = \bar{B}(\Delta_{p,u}) + \tilde{B}_u$ where $\tilde{B}_u := B_u - \bar{B}(\Delta_{p,u})$ satisfies $\mathbb{E}_u[\tilde{B}_u \mid \Delta_{p,u}] = 0$. Substituting into $\mathbb{E}_u[R_{p,u}]$:

$$\begin{aligned} \mathbb{E}_u\left[\frac{B_u \Delta_{p,u}}{p}\right] &= \underbrace{\mathbb{E}_u\left[\frac{\bar{B}(\Delta_{p,u}) \Delta_{p,u}}{p}\right]}_{\text{bias: conditional boundary smoothness}} + \underbrace{\mathbb{E}_u\left[\frac{\tilde{B}_u \Delta_{p,u}}{p}\right]}_{\text{noise: tower property}}. \end{aligned}$$

The *noise term* is exactly zero at the population level by the tower

$$\text{property (conditioning on } \Delta_{p,u}\text{): } \mathbb{E}_u\left[\frac{\Delta_{p,u}}{p} \cdot \underbrace{\mathbb{E}[\tilde{B}_u \mid \Delta_{p,u}]}_{=0}\right] = 0.$$

The *bias term* is second-order. Using (4): $\left|\mathbb{E}_u\left[\frac{\bar{B}(\Delta_{p,u}) \Delta_{p,u}}{p}\right]\right| \leq \frac{1}{p} \mathbb{E}_u[|\Delta_{p,u}| \cdot M|\Delta_{p,u}|] = \frac{M}{p} \mathbb{E}_u[\Delta_{p,u}^2]$.

THEOREM A.3 (OVERLAP-SHIFT ALIGNMENT EXPANSION — POPULATION LEVEL). Under Assumptions 1 and 2,

$$\mathbb{E}_u[\mathcal{A}_{p,u}] = -\frac{1}{p} \mathbb{E}_u[\mathbb{E}_u(s_{p,u}^0) \Delta_{p,u}] + r_p, \quad |r_p| \leq \frac{M}{p} \mathbb{E}_u[\Delta_{p,u}^2]. \quad (5)$$

The remainder r_p is purely second-order in the overlap shift (the error is quadratic in $\Delta_{p,u}$). Equivalently, with the exact per-user coefficient $c_{p,u} = -\partial S_{p,u}/p$ from (1):

$$\mathbb{E}_u[\mathcal{A}_{p,u}] = \mathbb{E}_u[c_{p,u} \Delta_{p,u}]$$

with zero remainder, since (1) is an identity.

PROOF. From (3), $\mathcal{A}_{p,u} = -\frac{\mathbb{E}_u(s_{p,u}^0)}{p} \Delta_{p,u} + R_{p,u}$ where $R_{p,u} = -\frac{B_u \Delta_{p,u}}{p}$. Set $r_p = \mathbb{E}_u[R_{p,u}]$. By the bias-noise decomposition above, the noise in B_u contributes zero to the population mean. The bias is bounded by $M \mathbb{E}_u[\Delta_{p,u}^2]/p$ as shown above. $\square$

For the baseline candidate set, $|C_u^0 \cap X_{p,u}^0| = p$, hence $\mathcal{A}_{p,u}(C_u^0) = 0$. Therefore the experimental alignment shift satisfies $\Delta \mathcal{A}_{p,u} = \mathcal{A}_{p,u}(C_u^E)$. Since $O_{p,u}^0 = p$ and $O_{p,u}^E = \hat{p}_u$, we have $\Delta_{p,u} = O_{p,u}^E - O_{p,u}^0$, which gives the overlap form invoked as Step 3 of Theorem 3.1 in the main text.

Remark 1 (Finite sample: the CLT term). In practice we observe n users and estimate $\mathbb{E}_u[\mathcal{A}_{p,u}]$ by the sample mean

$$\hat{\mathcal{A}}_p := \frac{1}{n} \sum_{u=1}^n \mathcal{A}_{p,u}.$$

The noise term $\tilde{B}_u$ that vanished at the population level reappears as sampling noise:

$$\hat{\mathcal{A}}_p = -\frac{1}{p} \cdot \frac{1}{n} \sum_u \mathbb{E}_u(s_{p,u}^0) \Delta_{p,u} + \underbrace{r_p}_{\text{bias } O(M \mathbb{E}[\Delta^2]/p)} + \underbrace{\varepsilon_n}_{\text{CLT noise } O_P(1/(p\sqrt{n}))},$$

where ε_n has mean zero with standard deviation $O(1/(p\sqrt{n}))$. Indeed, since $|B_u| \leq 1$, we have $|R_{p,u}| \leq |\Delta_{p,u}|/p$, so $\text{Var}(R_{p,u}) \leq \mathbb{E}[\Delta^2]/p^2$, and the sample mean concentrates at rate $1/\sqrt{n}$. The two error terms have different natures:

- **Bias** r_p : systematic, from conditional boundary smoothness. Shrinks as overlap shifts shrink. Independent of sample size.
- **CLT noise** ε_n : from per-user variance in B_u . Shrinks as $1/\sqrt{n}$; independent of overlap shifts and negligible when n is in the millions.

Remark 2 (Overlap is the unique first-order proxy). The exact identity (1) shows that $\mathcal{A}_{p,u}$ depends on the candidate set C_u through two quantities: the overlap shift $\Delta_{p,u}$ and the boundary engagement $\partial S_{p,u}$. The boundary engagement enters as a coefficient, while the candidate-set perturbation enters through $\Delta_{p,u} = O_{p,u}^E - p$. Thus overlap is the first-order selection coordinate for alignment. After aggregating prefixes, calibration learns the unknown coefficients, so the linear span of overlap metrics matches the first-order theory-based alignment space up to the second-order residual in Theorem A.3.

A.3 From full-pool overlap to logged overlap

The practical alignment metric (7) measures L_1 - L_2 ranking agreement within the logged preranker output $C(u)$, whereas the theoretical alignment channel (Step 3 of Theorem 3.1 in the main text) depends on the overlap $|C_u^E \cap X_{p,u}^0|$ defined over the full retrieval pool $D(u)$. This subsection bridges the two through a two-step composition:

- (1) *Step 1 (visibility)*. Restricting the full-pool threshold set $X_{p,u}^0$ to the logged production output C_{eval} produces a *visible-threshold overlap* $V_{p,u}^m$, governed by the production coverage π .
- (2) *Step 2 (effective prefix)*. Comparing top- q sets inside C_{eval} rather than counting items above a fixed full-pool threshold

rescales prefixes by approximately $1/\gamma$ and mixes neighboring prefixes.

Both steps are properties of the production logging distribution and the calibrated model family, so calibration absorbs their leading linear effect; the residual is second-order in the overlap shifts.

A.3.1 Full-pool, visible, and logged overlaps. Fix a request u (suppressed below). Let $C_{\text{eval}} \subseteq D$ denote the evaluation set—the candidate set selected by the production model—with $|C_{\text{eval}}| = k$. L_2 scores are available on C_{eval} .

- **Theoretical overlap:** $O_p^m := |C^m \cap X_p^0|$, with
 $C^m = \text{top}_k(D; L_1^m)$, $X_p^0 = \{e \in D : L_2(e) \geq L_2(s_p^0)\}$.
- **Visible threshold overlap:** $V_p^m := \sum_{e \in C^m \cap X_p^0} \pi(e)$, the expected number of theoretical-threshold items visible in the logged output.
- **Practical metric:** $\hat{O}_p^m := |\text{top}_p(C_{\text{eval}}; L_1^m) \cap \text{top}_p(C_{\text{eval}}; L_2)|$.

For each item $e \in D$, let $\pi(e) := P(e \in C_{\text{eval}} \mid u, D, L_2)$ denote the production model's selection probability—a property of the production model alone. Define the sets of above-threshold items that differ between prerankers:

- $A := (C^E \setminus C^0) \cap X_p^0$: above-threshold items the experiment adds.
- $B := (C^0 \setminus C^E) \cap X_p^0$: above-threshold items the experiment drops.

Items in $(C^0 \cap C^E) \cap X_p^0$ cancel when taking differences. The overlap shift satisfies $\Delta_p = |A| - |B|$.

Define the average coverage and centered deviation:

$$\bar{\pi} := \frac{1}{|X_p^0|} \sum_{e \in X_p^0} \pi(e), \quad \tilde{\pi}(e) := \pi(e) - \bar{\pi}.$$

A.3.2 Coverage-weighted overlap split. The visible threshold overlap is:

$$V_p^m := \sum_{e \in C^m \cap X_p^0} \pi(e) = \mathbb{E}_{C_{\text{eval}}} [|C^m \cap X_p^0 \cap C_{\text{eval}}|].$$

Since shared items cancel:

PROPOSITION A.4 (EXACT COVERAGE-WEIGHTED OVERLAP SPLIT).

$$V_p^E - V_p^0 = \bar{\pi} \cdot \Delta_p + \varepsilon, \quad (6)$$

$$\varepsilon := \sum_{e \in A} \tilde{\pi}(e) - \sum_{e \in B} \tilde{\pi}(e) = |A|(\bar{\pi}_A - \bar{\pi}) - |B|(\bar{\pi}_B - \bar{\pi}),$$

where $\bar{\pi}_A := \frac{1}{|A|} \sum_{e \in A} \pi(e)$ and $\bar{\pi}_B := \frac{1}{|B|} \sum_{e \in B} \pi(e)$. If A or B is empty, the corresponding product term is interpreted as zero; the sum expression for ε is the primary definition. This identity is exact.

PROOF. $\sum_{e \in A} \pi(e) - \sum_{e \in B} \pi(e) = \sum_{e \in A} [\bar{\pi} + \tilde{\pi}(e)] - \sum_{e \in B} [\bar{\pi} + \tilde{\pi}(e)] = \bar{\pi}(|A| - |B|) + \sum_{e \in A} \tilde{\pi}(e) - \sum_{e \in B} \tilde{\pi}(e) = \bar{\pi} \Delta_p + \varepsilon$. $\square$

The proportional term $\bar{\pi} \Delta_p$ is the prefix-local coverage contribution. The residual $\varepsilon = |A|(\bar{\pi}_A - \bar{\pi}) - |B|(\bar{\pi}_B - \bar{\pi})$ is the part of visible overlap not explained by the average production coverage $\bar{\pi}$. It is a product of experimental factors ($|A|$, $|B|$) and production-model factors ($\bar{\pi}_A - \bar{\pi}$, $\bar{\pi}_B - \bar{\pi}$).

A.3.3 The coverage residual is a selection covariance. We restore the user subscript u from here on, since the next two subsections average across users. At the per-user level, the quantities introduced in Section A.3 acquire a u subscript: $\varepsilon_u := \varepsilon$, $\Delta_{p,u} := \Delta_p$, $A_u := A$, $B_u := B$, $\tilde{\pi}_u := \tilde{\pi}$, and so on. The coverage residual ε_u has a revealing algebraic structure that does not require any assumption.

For each item $e \in X_p^0$, define the *centered selection indicator*:

$$\tilde{d}_u(e) := d_u(e) - \frac{\Delta_{p,u}}{|X_p^0|}, \quad \text{where } d_u(e) := \mathbf{1}[e \in A_u] - \mathbf{1}[e \in B_u].$$

Note that $\sum_{e \in X_p^0} \tilde{d}_u(e) = 0$ (centered) and $\sum_{e \in X_p^0} \tilde{\pi}_u(e) = 0$ (by definition).

LEMMA A.5 (COVERAGE-SELECTION COVARIANCE IDENTITY). The coverage residual ε_u equals the inner product of two centered vectors:

$$\varepsilon_u = \sum_{e \in X_p^0} \tilde{\pi}_u(e) \cdot \tilde{d}_u(e) = |X_p^0| \cdot \widehat{\text{Cov}}_u(\pi, d), \quad (7)$$

where $\widehat{\text{Cov}}_u(\pi, d) := \frac{1}{|X_p^0|} \sum_{e \in X_p^0} \tilde{\pi}_u(e) \cdot \tilde{d}_u(e)$ is the sample covariance between coverage and selection over the threshold set.

PROOF.

$$\begin{aligned} \sum_e \tilde{\pi}_u(e) \cdot d_u(e) &= \sum_e \tilde{\pi}_u(e) [\tilde{d}_u(e) + \Delta_{p,u}/|X_p^0|] \\ &= \sum_e \tilde{\pi}_u(e) \cdot \tilde{d}_u(e) + \frac{\Delta_{p,u}}{|X_p^0|} \underbrace{\sum_e \tilde{\pi}_u(e)}_{=0} \\ &= \sum_e \tilde{\pi}_u(e) \cdot \tilde{d}_u(e). \end{aligned}$$

Separately, $\sum_e \tilde{\pi}_u(e) \cdot d_u(e) = \sum_{e \in A} \tilde{\pi}_u(e) - \sum_{e \in B} \tilde{\pi}_u(e) = \varepsilon_u$. $\square$

Equation (7) is exact and assumption-free. It reveals the “ $\delta \times \delta$ ” structure concretely. Here $\tilde{\pi}_u(e)$ is the *coverage deviation* (how the production model's coverage of item e deviates from the threshold-set average), while $\tilde{d}_u(e)$ is the *centered selection shift* (whether the experiment adds or drops item e , centered so the mean is zero). The coverage residual is their inner product: it is large only when the two vectors are aligned across items.

A.3.4 Step 1: Full-pool to visible-overlap bridge. We now move from the per-user coverage residual $\varepsilon_{p,u}$ to its population-level expectation, following the same bias-noise strategy used for the alignment channel (Theorem A.3). Define the *conditional expected coverage error*:

$$\bar{\varepsilon}(d) := \mathbb{E}_u[\varepsilon_u \mid \Delta_{p,u} = d],$$

and decompose ε_u into its conditional mean and a centered residual:

$$\varepsilon_u = \bar{\varepsilon}(\Delta_{p,u}) + \tilde{\varepsilon}_u, \quad \tilde{\varepsilon}_u := \varepsilon_u - \bar{\varepsilon}(\Delta_{p,u}), \quad \mathbb{E}_u[\tilde{\varepsilon}_u \mid \Delta_{p,u}] = 0.$$

By the tower property, $\mathbb{E}_u[\tilde{\varepsilon}_u] = 0$, so $\mathbb{E}_u[\varepsilon_u] = \mathbb{E}_u[\bar{\varepsilon}(\Delta_{p,u})]$ and only the systematic part $\bar{\varepsilon}$ requires a stability assumption.

Assumption 3 (Stable full-pool to visible-overlap bridge). The bridge from full-pool overlap shifts $\Delta O_{p,u}$ to visible-overlap differences $V_{p,u}^E - V_{p,u}^0$ has two components, each governed by a *rule* that depends on the production logging policy and prefix p but not on which calibrated pair (L_1^E, L_1^0) is being analyzed.

(i) *Smoothness of the conditional residual coverage error.* Define the conditional expected coverage error as a real-valued function of the overlap shift,

$$\bar{\varepsilon}_p(d) := \mathbb{E}_u[\varepsilon_{p,u} \mid \Delta O_{p,u} = d],$$

where $\varepsilon_{p,u}$ is the coverage residual from Proposition A.4. We assume $\bar{\varepsilon}_p$ admits a Taylor expansion around $d = 0$ with quadratic remainder,

$$\bar{\varepsilon}_p(d) = c_p^V + \delta_p^V d + \rho_p^V(d), \quad |\rho_p^V(d)| \leq \frac{M_\pi^V}{2} d^2,$$

where the constants c_p^V , δ_p^V , and M_π^V are properties of the production logging policy and the prefix p , not of the calibrated pair.

(ii) *Stability of the coverage-shift covariance.* The user-averaged product of coverage and overlap shift admits a first-order linear representation in the vector of overlap shifts across prefixes,

$$\begin{aligned} \mathbb{E}_u[\bar{\pi}_{p,u} \Delta O_{p,u}] &= \sum_{r=1}^k \gamma_{pr} \mathbb{E}_u[\Delta O_{r,u}] + r_p^\pi, \\ |r_p^\pi| &\leq C^\pi \sum_{r=1}^k \mathbb{E}_u[\Delta O_{r,u}^2]. \end{aligned}$$

Again, γ_{pr} and C^π depend on the production logging policy and prefix grid but not on the calibrated pair.

What “pair-independent” allows and what it forbids. The assumption does *not* require different pairs to produce the same overlap-shift distributions, the same swap volumes, or the same correlations with the production model—those are precisely the differences the calibrated metric M_{align} is designed to measure. What it requires is that the *rule* converting an overlap shift d into the expected residual coverage error $\bar{\varepsilon}_p(d)$ has the same Taylor coefficients across pairs. Intuitively, c_p^V , δ_p^V , and γ_{pr} are determined by the joint distribution of (production coverage $\pi(e)$, L_2 score, item identity) over the retrieval pool D , all of which are properties of the production logging policy held fixed across the calibrated family.

Why correlation drift between pairs does not invalidate the assumption. A natural concern is that an experimental preranker may have a different correlation with the production model than the baseline does—write $\eta := \rho^E - \rho^0$ —and that this drift is itself first-order in the size of the experimental change. Concretely, the swap-asymmetry component of ε_u can be written as

$$\varepsilon_u = \frac{S_u}{2} (\bar{\pi}_A^u - \bar{\pi}_B^u) + \frac{\Delta O_{p,u}}{2} (\bar{\pi}_A^u + \bar{\pi}_B^u - 2\bar{\pi}^u), \quad S_u := |A_u| + |B_u|,$$

and the drift η shifts $\bar{\pi}_A^u - \bar{\pi}_B^u$ by $O(\eta)$. At face value this could shift $\bar{\varepsilon}_p(d)$ pointwise by $O(\eta)$, violating clause (i)’s pair-independence. The resolution is that, for the calibrated family, η varies across pairs in a way that correlates with the pair’s overlap-shift magnitude—more aggressive prerankers tend to shift both quantities together—so the pair-to-pair perturbation of $\bar{\varepsilon}_p$ enters the regression as a contribution *proportional to the regressor* M_{align} and is absorbed by the fitted slope α . The assumption is therefore best read as “shared up to a pair-specific perturbation that the regression slope absorbs.” What it strictly forbids is pair-to-pair variation orthogonal to $(M_{\text{align}}, M_{\text{acc}})$ —for instance, a model architecture whose correlation drift is anomalously large relative to its overlap-shift magnitude. Such variation would create omitted-variable bias that

the linear regression cannot absorb, which is why the calibration must be refit whenever the production model, retrieval L_0 , or model architecture family changes.

THEOREM A.6 (FULL-POOL TO VISIBLE-OVERLAP BRIDGE). *Under Assumption 3, there exists a production-dependent matrix $K^V = (K_{qp}^V)$ and constants c_q^V such that*

$$\mathbb{E}_u[V_{q,u}^E - V_{q,u}^0] = \sum_{p=1}^k K_{qp}^V \mathbb{E}_u[\Delta O_{p,u}] + c_q^V + r_q^V, \quad (8)$$

with

$$|r_q^V| \leq C^\pi \sum_{p=1}^k \mathbb{E}_u[\Delta O_{p,u}^2].$$

Here $\Delta O_{p,u} := O_{p,u}^E - O_{p,u}^0$. In the special case where coverage is prefix-local and independent of the overlap shift, K^V is diagonal and K_{pp}^V reduces to the scalar coverage factor $\gamma_p + \delta_p^V$.

PROOF. For a fixed prefix p , Proposition A.4 gives the per-user identity

$$V_{p,u}^E - V_{p,u}^0 = \bar{\pi}_{p,u} \Delta O_{p,u} + \varepsilon_{p,u}.$$

Averaging over users:

$$\mathbb{E}_u[V_p^E - V_p^0] = \mathbb{E}_u[\bar{\pi}_{p,u} \Delta O_{p,u}] + \mathbb{E}_u[\varepsilon_{p,u}].$$

Coverage covariance term. Assumption 3.2 expands

$$\mathbb{E}_u[\bar{\pi}_{p,u} \Delta O_{p,u}] = \sum_r \gamma_{pr} \mathbb{E}_u[\Delta O_{r,u}] + r_p^\pi,$$

with $|r_p^\pi| \leq C^\pi \sum_r \mathbb{E}_u[\Delta O_{r,u}^2]$.

Conditional residual term. By the tower property, $\mathbb{E}_u[\varepsilon_{p,u}] = \mathbb{E}_u[\bar{\varepsilon}_p(\Delta O_{p,u})]$. Substituting Assumption 3.1 yields

$$\mathbb{E}_u[\bar{\varepsilon}_p(\Delta O_{p,u})] = c_p^V + \delta_p^V \mathbb{E}_u[\Delta O_{p,u}] + \mathbb{E}_u[\rho_p^V(\Delta O_{p,u})],$$

with $|\mathbb{E}_u[\rho_p^V(\Delta O_{p,u})]| \leq \frac{M_\pi^V}{2} \mathbb{E}_u[\Delta O_{p,u}^2]$.

Combining. Set $K_{qp}^V := \gamma_{qp} + \delta_q^V \mathbf{1}[p = q]$ and $r_q^V := r_q^\pi + \mathbb{E}_u[\rho_q^V(\Delta O_{q,u})]$. Then

$$\begin{aligned} |r_q^V| &\leq C^\pi \sum_p \mathbb{E}_u[\Delta O_{p,u}^2] + \frac{M_\pi^V}{2} \mathbb{E}_u[\Delta O_{q,u}^2] \\ &\leq C^\pi \sum_p \mathbb{E}_u[\Delta O_{p,u}^2], \end{aligned}$$

with $C^\pi := C^\pi + M_\pi^V/2$. Stacking prefixes gives (8). $\square$

A.3.5 Step 2: Visible-overlap to logged-overlap map. Theorem A.6 relates the full-pool overlap to the visible-threshold overlap V_p^m , which still references the full-pool threshold set X_p^0 . The implemented metric $\hat{O}_q^m$ instead compares two top- q sets inside C_{eval} . If the production model covers a fraction γ of the theoretical threshold region, then the top- q set inside C_{eval} corresponds roughly to a full-pool threshold near q/γ . When coverage varies by user and rank, this correspondence is not a scalar rescaling but a linear mixing of neighboring theoretical prefixes.

Assumption 4 (Stable effective-prefix map). For the calibrated model family, the expected logged top-prefix overlap is a stable

linear function of the expected visible-threshold overlaps up to a second-order residual:

$$\mathbb{E}_u[\Delta\hat{O}_{q,u}] = \sum_{r=1}^k K_{qr}^H \mathbb{E}_u[V_{r,u}^E - V_{r,u}^0] + c_q^H + r_q^H,$$

$$|r_q^H| \leq C_\pi^H \sum_{p=1}^k \mathbb{E}_u[\Delta O_{p,u}^2].$$

The operator K^H and constants c_q^H depend on the production logging distribution and prefix grid, but are stable across model pairs in calibration. We further assume K^H is row-bounded: $\sum_r |K_{qr}^H| \leq K_{\max}^H < \infty$.

THEOREM A.7 (VISIBLE-OVERLAP TO LOGGED-OVERLAP BRIDGE). *Under Assumptions 3 and 4, there exists a production-dependent linear operator $K = K^H K^V$ and constants c_q such that*

$$\mathbb{E}_u[\Delta\hat{O}_{q,u}] = \sum_{p=1}^k K_{qp} \mathbb{E}_u[\Delta O_{p,u}] + c_q + r_q^{\text{cov}}, \quad (9)$$

with

$$|r_q^{\text{cov}}| \leq C_\pi \sum_{p=1}^k \mathbb{E}_u[\Delta O_{p,u}^2].$$

The operator K induces the effective alignment weights learned by calibration; the additive constants c_q vanish under the symmetric-retraining design used in our calibration pipeline (§D), so the calibration is fit without an intercept.

PROOF. Substitute the visible-overlap representation (8) into Assumption 4:

$$\begin{aligned} \mathbb{E}_u[\Delta\hat{O}_{q,u}] &= \sum_r K_{qr}^H \left[\sum_p K_{rp}^V \mathbb{E}_u[\Delta O_{p,u}] + c_r^V + r_r^V \right] \\ &\quad + c_q^H + r_q^H \\ &= \sum_p (K^H K^V)_{qp} \mathbb{E}_u[\Delta O_{p,u}] + c_q + r_q^{\text{cov}}, \end{aligned}$$

where

$$\begin{aligned} (K^H K^V)_{qp} &:= \sum_r K_{qr}^H K_{rp}^V, \\ c_q &:= \sum_r K_{qr}^H c_r^V + c_q^H, \\ r_q^{\text{cov}} &:= \sum_r K_{qr}^H r_r^V + r_q^H. \end{aligned}$$

Setting $K := K^H K^V$ gives the stated linear form. The residual bound follows by the triangle inequality and Assumption 4's row-bound on K^H :

$$|r_q^{\text{cov}}| \leq \sum_r |K_{qr}^H| \cdot |r_r^V| + |r_q^H| \leq (K_{\max}^H C_\pi^V + C_\pi^H) \sum_{p=1}^k \mathbb{E}_u[\Delta O_{p,u}^2],$$

so $C_\pi := K_{\max}^H C_\pi^V + C_\pi^H$ works. $\square$

The structural parallel with the alignment proof is exact: both use (i) bias-noise decomposition via the tower property, (ii) a stable conditional mean assumption, and (iii) second-order residuals in the overlap shifts.

A.3.6 Resolution limit and the pipeline stage. The two-step bridge above shows that the practical metric is calibratable, but it does not yet reveal *when* the metric carries useful information. This subsection unpacks the geometric meaning of the operator K and identifies the regime where the metric resolves alignment differences.

Effective prefix. The theoretical threshold set X_p^0 contains p items from D , of which an expected γp appear in C_{eval} . Meanwhile, the practical set $\hat{X}_p := \text{top}_p(C_{\text{eval}}; L_2)$ contains exactly p items—the γp items from $X_p^0 \cap C_{\text{eval}}$ plus $(1 - \gamma)p$ additional items that lie below the theoretical threshold but rank in the top p of C_{eval} . Inverting: $\hat{X}_p$ spans roughly the same items as $X_{p/\gamma}^0 \cap C_{\text{eval}}$ —the practical cutoff p in C_{eval} corresponds to a larger theoretical cutoff p/γ in D . This is the “effective prefix” interpretation: restricting to the smaller pool C_{eval} (size k) inflates the effective prefix by $1/\gamma$.

The metric remains a valid alignment proxy because:

- (1) The effective-prefix operator is stable for the calibrated model family—it is mainly a property of the production logging distribution.
- (2) The calibration coefficient α absorbs the induced linear combination of theoretical reward weights.
- (3) When coverage is approximately monotone in L_2 quality, a model that improves theoretical overlap across neighboring prefixes also improves practical overlap after the same linear mixing.

Two production-coverage extremes.

- **Production model = L_2 (perfect coverage):** $X_p^0 \subseteq C_{\text{eval}}$, so $\gamma \approx 1$.
- **Production model = random:** each item appears with probability $k/|D|$, giving $\gamma \approx k/|D|$ —the same for all models by symmetry.

Resolution limit. The effective-prefix interpretation reveals a fundamental constraint: the inflated prefix p/γ must not exceed k (the pool size), otherwise the practical metric at prefix p is trying to measure alignment at a quality level coarser than the preranker's entire output. The metric at prefix p is informative only when

$$\gamma \geq \frac{p}{k}, \quad (10)$$

i.e., when C_{eval} contains enough items from the theoretical threshold set X_p^0 to distinguish between models. When $\gamma p \lesssim 1$, virtually no items from X_p^0 are visible in C_{eval} , and the practical overlap at prefix p is dominated by items below the theoretical threshold—it measures noise rather than alignment.

Why alignment must be measured on the preranker output. The three-stage pipeline $D \xrightarrow{L_1} C_{\text{eval}} \xrightarrow{L_2} I$ creates three candidate pools, and the alignment metric can only live on the middle one:

- **Upstream: the retrieval pool D (size n).** Computing overlap on D would require running L_2 on all n items—precisely the computation the preranker exists to avoid.
- **Downstream: the impressed set I (L_2 's output, size $|I| \ll k$).** The production model, the baseline L_1^0 , and the experiment L_1^E all operate at the same cutoff k . Items in I have already survived both L_1^{prod} selection and L_2 re-ranking, so measuring

overlap within I compares models only on items they all agree are good—the metric loses discriminative power because the “bad” items have been filtered out by the shared L_2 stage. Furthermore, $|I|$ is typically 10–50 per user, making the inflated prefix p/γ_I exceed the pool size at virtually every meaningful prefix.

- *The preranker output C_{eval} (size k , typically hundreds).* This is the unique set where both L_1 and L_2 scores are available: L_2 has scored every item in C_{eval} (it is L_2 's input), and we can re-score these items with any candidate L_1 model. The pool is large enough to resolve the L_2 quality ranking at relevant prefixes.

C_{eval} is therefore not merely a convenient choice but the only stage in the pipeline where the alignment metric is both cheaply computable and informative. The accuracy channel, by contrast, is a *statistical* estimator (Theorem A.8) that converges by the law of large numbers across users; it can use engagement labels from I because per-user signal aggregates well. The alignment channel is *algebraic* and requires a pool large enough to resolve the L_2 quality ranking, which only C_{eval} provides.

Scaling with production model quality.

- *Good production model* (γ close to 1): the condition (10) is satisfied for all $p \leq k$, and the practical metric has full resolution.
- *Mediocre production model* ($\gamma \approx k/n$, comparable to random selection): the condition becomes $p \leq k^2/n$. For small prefixes (high positions) the metric still works; for large prefixes it loses signal because C_{eval} rarely contains items from the relevant quality range.
- *Practical implication:* the production model used to generate C_{eval} must itself have reasonable alignment with L_2 . Evaluating experimental prerankers against a production model that is far from L_2 degrades the alignment metric at all but the smallest prefixes.

A.3.7 *What calibration absorbs.* In the combined offline score $\widehat{M} = \alpha M_{\text{align}} + \beta M_{\text{acc}}$:

- The slope α absorbs the effective alignment weights induced by the operator $K = K^H K^V$ after aggregating prefixes into the scalar metric M_{align} .
- The individual logged-overlap metrics need not estimate individual theoretical prefixes; their calibrated linear span represents the same first-order overlap space after applying the stable operator K .
- The constants c_q vanish by the symmetric-retraining design (§D), so no intercept is needed.
- The residual calibration error is $O(C_\pi \sum_p \mathbb{E}[\Delta O_p^2])$, which vanishes as the experimental perturbation shrinks.

Under full independence and uniform coverage, K reduces to a diagonal scalar coverage correction and the constants vanish; under realistic user- and rank-dependent coverage, K mixes neighboring prefixes but remains stable across the calibrated family (Assumption 3, Assumption 4).

A.4 Impressed-accuracy estimator consistency

For the accuracy term we work in a population view across requests. Let $u_1, \dots, u_n$ denote i.i.d. requests, and for each request u_t , let D_t be its candidate set. For each t , define the baseline threshold set

$$X_{p,t}^0 := \{e \in D_t : L_2(e) \geq L_2(s_{p,t}^0)\},$$

and the corresponding preranked candidates C_t^0 and C_t^E . Let $Y_t(e) \in [0, 1]$ denote the observed engagement label for item e under request u_t , and assume $\mathbb{E}[Y_t(e) \mid u_t, e] = \mu_t(e)$.

The per-request accuracy estimand is

$$A_{p,t} := \frac{\sum_{e \in C_t^E \cap X_{p,t}^0} \mu_t(e)}{|C_t^E \cap X_{p,t}^0|} - \frac{\sum_{e \in C_t^0 \cap X_{p,t}^0} \mu_t(e)}{|C_t^0 \cap X_{p,t}^0|},$$

and the population accuracy is $\text{Acc}_p := \mathbb{E}[A_{p,t}]$.

We estimate it using the plugin estimator

$$\widehat{\text{Acc}}_p = \frac{1}{n} \sum_{t=1}^n \left(\frac{\sum_{e \in C_t^E \cap X_{p,t}^0} Y_t(e)}{|C_t^E \cap X_{p,t}^0|} - \frac{\sum_{e \in C_t^0 \cap X_{p,t}^0} Y_t(e)}{|C_t^0 \cap X_{p,t}^0|} \right).$$

Assumption 5 (I.i.d. requests and unbiased engagement labels).

The requests $u_1, \dots, u_n$ are i.i.d. For each request u_t :

- $\mathbb{E}[Y_t(e) \mid u_t, e] = \mu_t(e)$;
- $Y_t(e)$ is uniformly bounded;
- for both $m \in \{0, E\}$, $|C_t^m \cap X_{p,t}^0|$ is almost surely nonzero.

THEOREM A.8 (IMPRESSED-ACCURACY ESTIMATOR CONSISTENCY AND ASYMPTOTIC NORMALITY). *Under Assumption 5,*

$$\widehat{\text{Acc}}_p \xrightarrow{a.s.} \text{Acc}_p$$

as $n \rightarrow \infty$. Moreover, if the per-request variance $\text{Var}(Z_{p,t})$ is finite, then

$$\sqrt{n}(\widehat{\text{Acc}}_p - \text{Acc}_p) \xrightarrow{d} \mathcal{N}(0, \sigma_p^2)$$

for some $\sigma_p^2 \geq 0$.

PROOF. Define $Z_{p,t} := \frac{\sum_{e \in C_t^E \cap X_{p,t}^0} Y_t(e)}{|C_t^E \cap X_{p,t}^0|} - \frac{\sum_{e \in C_t^0 \cap X_{p,t}^0} Y_t(e)}{|C_t^0 \cap X_{p,t}^0|}$. Then

$\widehat{\text{Acc}}_p = \frac{1}{n} \sum_{t=1}^n Z_{p,t}$. By conditional unbiasedness, $\mathbb{E}[Z_{p,t} \mid u_t] = A_{p,t}$, so $\mathbb{E}[Z_{p,t}] = \text{Acc}_p$. Since $Z_{p,t}$ is bounded and i.i.d. under Assumption 5, Kolmogorov's strong law of large numbers gives almost sure convergence; the Lindeberg–Lévy central limit theorem gives asymptotic normality when the variance is finite. $\square$

Theorem A.8 justifies using $\widehat{\text{Acc}}_p$ (and its normalized variant $N_p^E - N_p^0$) as a statistically valid normalized accuracy proxy at prefix p . The theory-based channel $\mathcal{B}_{p,u}$ uses the fixed denominator p , while this proxy normalizes by the number of above-threshold selected items. The calibrated coefficient β_p absorbs this scale difference, so the claim is again about the calibrated linear span of accuracy metrics.

A.5 Cascade Reward Representation: per-prefix form

Combining the overlap-shift alignment expansion (Theorem A.3), the full-pool to logged-overlap bridge (Theorem A.7), and the consistency of the normalized impressed-accuracy estimator (Theorem A.8) gives the population-level reward-lift expression in terms

of the metrics actually computed—the per-prefix form of the main paper’s Cascade Reward Representation (Theorem 3.1).

THEOREM A.9 (CASCADE REWARD REPRESENTATION, PER-PREFIX POPULATION FORM). *This theorem is the per-prefix population form of the Cascade Reward Representation (Theorem 3.1 in the main text), before scalar aggregation. Under Assumptions 1–5, the expected reward lift admits the population-level decomposition*

$$\mathbb{E}_u[\mathbf{R}_u^E - \mathbf{R}_u^0] = \sum_{p=1}^k \tilde{\alpha}_p \mathbb{E}_u[\Delta \hat{O}_{p,u}] + \sum_{p=1}^k \beta_p \mathbb{E}_u[\Delta N_{p,u}] + \mathcal{R},$$

where $\tilde{\alpha}_p := -\frac{1}{p} \sum_{q=1}^k w_p \mathbb{E}_u[\mathbf{E}_u(s_{q,u}^0)] K_{qp}^{-1}$ are effective alignment weights induced by the logged-overlap operator K from Theorem A.7, $\beta_p := w_p$ are the position weights from (3) acting as accuracy weights, and the remainder

$$\mathcal{R} = \underbrace{\sum_p w_p r_p}_{\text{alignment bias } O(M \mathbb{E}[\Delta^2]/p)} + \underbrace{\sum_p \tilde{\alpha}_p r_p^{\text{cov}}}_{\text{logged-overlap coverage residual } O(C_\pi \sum_p \mathbb{E}[\Delta O_p^2])} + \underbrace{\varepsilon_n}_{\text{accuracy noise } O_p(1/\sqrt{n})}$$

combines the alignment first-order remainder, the logged-overlap coverage residual, and the CLT noise of the accuracy estimator.

PROOF. Apply the prefix-average expansion (3) from §A.1 to express the reward lift as

$$\sum_p w_p (\mathbb{E}_u[\mathcal{A}_{p,u}] + \mathbb{E}_u[\Delta \mathcal{B}_{p,u}]).$$

Substitute Theorem A.3 for $\mathbb{E}_u[\mathcal{A}_{p,u}]$, expressing it as a linear combination of $\mathbb{E}_u[\Delta_{p,u}]$ with second-order remainder. Theorem A.7 then converts $\mathbb{E}_u[\Delta_{p,u}]$ into a linear combination of $\mathbb{E}_u[\Delta \hat{O}_{q,u}]$ via the operator K , contributing the coverage-error remainder. Under the symmetric-retraining design (§D), the additive constants c_q produced by Theorem A.7 vanish. Theorem A.8 estimates $\mathbb{E}_u[\Delta \mathcal{B}_{p,u}]$ by $\mathbb{E}_u[\Delta N_{p,u}]$ with $O_p(1/\sqrt{n})$ error. Collecting terms gives the stated decomposition. $\square$

The calibration coefficients α, β in $\hat{M} = \alpha M_{\text{align}} + \beta M_{\text{acc}}$ absorb the unknown $\tilde{\alpha}_p, \beta_p$ once the per-prefix metrics are aggregated to the scalars $M_{\text{align}}(\rho)$ and M_{acc} , with the further co-monotonicity assumption discussed in the body.

B Dual-Distribution Training Infrastructure

The alignment–accuracy training objective $L = \lambda L_{\text{align}} + (1-\lambda) L_{\text{acc}}$ (Section 4.3) requires consuming two data streams in each training step: impressed examples for L_{acc} and unimpressed candidate-pool examples for L_{align} [4]. We use a *trainer-side Ray union* [5] of two independent dataloaders rather than a workflow-level union that would merge the streams before the trainer. The advantage is flexibility: the mixing ratio between the two datasets is controlled by an `unimpressed_ratio` hyperparameter at training time rather than being hard-coded into the dataset definition.

Dataloader throughput mismatch. Each Ray dataloader has an effective loading speed determined largely by its dataset layout (shard sizes, storage bandwidth), while the consumption rate from each stream is governed by `unimpressed_ratio`. Because loading and consumption are controlled by different factors, these rates rarely match exactly. The faster dataloader produces excess samples that accumulate in memory and can eventually cause out-of-memory (OOM) failures.

Handling excess batches. We considered several strategies for problematic batches from the over-producing dataloader:

- **Drop at the dataloader.** The dataloader idles while discarding, leading to timeouts when it falls too far behind.
- **Drop at inference.** Different GPU workers may discard different numbers of batches, creating an accumulating imbalance that causes GPU synchronization failures.
- **Pass through unchanged.** Skewed batches make the training loss unstable and prevent convergence.

We instead **zero out excess batches at the loss stage**: the batch flows through the forward pass normally, keeping all GPUs synchronized, but its loss contribution is multiplied by zero so that no gradients propagate. On average, approximately 40.6% of batches are zeroed out, so we run roughly twice as many iterations to compensate. The resulting training cost is approximately 16.5 hours per run (~1.8× the 9-hour single-distribution baseline in both wall-clock time and compute cost). Despite the overhead, this is our most efficient method for multi-distribution training given available infrastructure, balancing iteration speed with stable convergence.

C Efficient Pairwise Alignment Loss

The pairwise alignment loss $L_{\text{align-pair}}$ [6] (Section 4.2) sums over all item pairs (i, j) that share the same request within a minibatch. With a batch size of $B \approx 42,000$, the naïve $O(B^2)$ enumeration is prohibitively memory-intensive. Assigning each pair to a single row and accumulating gradients via scatter operations avoids the memory issue but introduces write collisions on the GPU, which defeats parallelism and results in slow training.

Per-request blocks make pair enumeration sparse. Each request contributes at most $k = 100$ items to the batch, so the number of valid pairs is at most $O(Bk)$, not $O(B^2)$. To exploit this structure while retaining GPU parallelism, we proceed as follows.

- (1) **Hash grouping.** Compute a hash identifier $h = \text{user_id} \oplus \text{chunk_id}$ for each item in the batch, representing its originating request.
- (2) **Sort by hash.** Sort the batch by h in $O(B \log B)$ time. After sorting, items from the same request form contiguous blocks along the batch index.
- (3) **Block-diagonal structure.** In the $B \times B$ pair matrix, same-request items now occupy contiguous blocks along the diagonal, each of size at most $k \times k$.
- (4) **Chunk the batch.** Partition the sorted batch into contiguous chunks of size k (handling boundary cases separately).
- (5) **Scan the near-diagonal region.** For each chunk, compute pairwise losses only within the chunk itself and with its immediate neighbors (one block above and one below the diagonal). Any pair (i, j) with $|i - j| \geq k$ that shares a hash

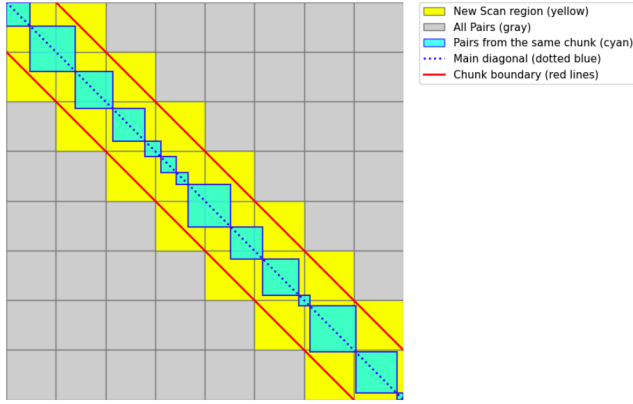


Figure 2: Block-diagonal structure of valid pairs after hash-based sorting. Cyan squares on the diagonal represent same-request item blocks (at most $k \times k$). The yellow band (diagonal $\pm$ one block) is the only region that can contain valid pairs; the grey region is provably empty and need not be scanned.

would require at least $|i - j| + 1 > k$ items with the same hash in the sorted order, contradicting the per-request bound. Therefore no valid pair lies outside this near-diagonal band, and we need not scan it.

The total cost is $O(Bk)$ for pair computation plus $O(B \log B)$ for sorting, giving $O(B(k + \log B))$ overall. All operations within each chunk are dense tensor computations that parallelize efficiently on GPU hardware, avoiding the scatter-based gradient collisions of the naïve approach. Figure 2 illustrates the block-diagonal structure after sorting.

D Additional Experimental Details

D.1 Offline calibration methodology

The offline metric calibration described in Section 5.1 uses positive (nonnegative-constrained) *no-intercept* linear regression to fit (α, β) from historical experiments. Each datapoint corresponds to a single model variant from an A/B test, with features $(M_{\text{align}}, M_{\text{acc}})$ and target Δy equal to the observed online lift in save engagement. The training set comprises 51 model variants drawn from 4 experiments run between July and December 2025. The forward-test set comprises 20 model variants from 3 experiments run between March and April 2026, ensuring a temporal gap of at least two months to test robustness to distribution shift.

Why no intercept. The baseline preranker L_1^0 is retrained on the same data window as the experimental preranker L_1^E . By Lemma A.5, the residual coverage error $\varepsilon_u = |X_u^0| \cdot \widehat{\text{Cov}}_u(\pi, d)$ vanishes in expectation when the centered swap indicator $d_u(e)$ has no systematic correlation with the production coverage $\pi(e)$, which holds when the two prerankers are trained on the same data and differ only in architecture or training objective. Consequently, the conditional residual mean $\bar{\varepsilon}_p(0)$ from Assumption 3 equals zero by construction, the constants c_p^V and c_q^H in Theorem A.7 reduce to zero, and the Cascade Reward Representation contains no constant term. We

therefore fit $\widehat{M} = \alpha M_{\text{align}} + \beta M_{\text{acc}}$ with no intercept; the fitted residual is consistent with this specification.

Why nonnegative slopes. Nonnegativity constraints on α and β are motivated by the Cascade Reward Representation, in which both calibrated coefficients are nonnegative under the standing assumptions. In practice, unconstrained regression yields similar coefficients (both are positive), but the constrained version provides a modest improvement in forward-test correlation, likely because it prevents overfitting to noise in the small training set.

D.2 Sensitivity to the overlap fraction ρ

Table 1 reports offline calibration results for the alignment metric at five values of the overlap fraction ρ (Section 4.1), using the Unimp. overlap method on the same 51-train / 20-test split as Section 5.1. Performance is stable for $\rho \in [0.2, 0.8]$: prediction accuracy remains at 85% and Pearson r varies only between 0.825 and 0.838. At $\rho=0.1$ both metrics degrade noticeably (prediction accuracy drops to 70%, Pearson r to 0.342). This is likely because a very small top fraction is noisy and dominated by tie-breaking effects in the L_1/L_2 orderings.

Table 1: Sensitivity of offline calibration to the overlap fraction ρ . Method: Unimp. overlap. Same train/test split as Table 1.

ρ	Pred. acc. (% , 95% CI)	Pearson r (95% CI)
0.1	70 [46, 88]	0.342 [-0.12, 0.68]
0.2	85 [60, 96]	0.838 [0.63, 0.94]
0.5	85 [60, 96]	0.838 [0.63, 0.94]
0.6	85 [60, 96]	0.835 [0.62, 0.94]
0.8	85 [60, 96]	0.825 [0.59, 0.93]

D.3 Engagement-weight tuning experiment

In addition to the five training configurations reported in Table 2, we conducted an additional experiment to test whether the proposed framework expands the feasible tradeoff frontier between deeper engagement (save) and lighter engagement (content view).

Setup. Starting from the proposed model T (accuracy loss + KL and pairwise alignment on unimpressed data), we tuned the multi-task head weights controlling the relative emphasis on save vs. content-view predictions. This is distinct from the alignment-accuracy mixing coefficient λ : it adjusts the *engagement-type* tradeoff within the accuracy branch, whereas λ controls the alignment-accuracy balance.

Results. Compared to the production baseline B2, the tuned variant achieved +1.12% content view with neutral save ($p > 0.05$). This complements the main result (T vs. B2: +0.62% save, -0.97% content view), demonstrating that the method does not merely shift the tradeoff curve but *expands the feasible operating region*: one can recover content-view performance without sacrificing the save gains that the alignment framework provides.

D.4 Experimental protocol

All A/B experiments follow the same protocol:

- **Surface:** homefeed (the primary content discovery surface).

- **Duration:** two weeks per experiment.
- **Traffic:** 2% of traffic per arm, randomly assigned at the user level.
- **Significance:** all reported non-neutral effects satisfy $p < 0.001$.
- **Guardrails:** weekly active users, hide rate, report rate, and other trust-and-safety metrics are monitored; all are neutral across all experiments.
- **Inference:** all arms share the same model architecture and serving infrastructure; only the training objective and data differ. No serving latency or cost changes.

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